

Movie Inputs

Budget (\$)

100000000.00

-

+

Popularity

20.00

-

+

Runtime (minutes)

120.00

-

+

Vote Average

7.00

Genre

Adventure

▼



MovieIQ - Predictive Analytics on Film Success

Analyze movie performance and predict potential revenue and success.



Movie Predictions

Predict Revenue

Predict Success

Prediction: Successful Movie

Success Probability

91.5%



Model Performance

Revenue Prediction

R² Score

0.5894

MAE

\$66.66M

RMSE

\$87.37M

Success Prediction

Accuracy

81.00%

Precision

80.95%

Recall

100.00%

F1 Score

89.47%



Business Recommendations

- **Horror** demonstrated the highest average ROI at approximately **83.26%**.
- Budget has a strong positive relationship with revenue (**0.76 correlation**), but higher spending does not guarantee success.
- Popularity has almost no correlation with revenue (**0.014**).
- Vote average has almost no correlation with revenue (**-0.005**).
- Revenue prediction can be used as an initial planning estimate, but not as a guaranteed outcome.
- The success classifier should be interpreted cautiously because it has limited ability to identify unsuccessful movies.

Note: The success model defines success as Revenue > Budget. Revenue-derived variables were excluded from the predictive features to avoid data leakage.